SigmaPay
Controller
output
input
I9output
void sendResponse(string userId, string responseData)
void displayError(string userId, string errorMessage)
I8input
void receiveUserRequest(string requestType, string requestData)
void captureUserAction(string userId, string actionType)
Business Services
IReportingService
+ String generateIncomeStatement(String userId, String period)
+ String generateMonthlySummary(String userId, String month)
+ boolean exportToPDF(String reportId)
+ IComparisonReport comparePeriods(String userId, String p1, String p2)
IBudgetService
+ String createBudget(String userId, double total, Date start, Date end, List<String> categories)
+ String recordExpense(String userId, double amount, String category, Date at, String method)
+ boolean sendAlertIfExceeded(String userId, String period)
+ ISpendingTrends getTrends(String userId, String period)
IGoalService
+ String setGoal(String userId, String type, double target, Date deadline, String priority)
+ IGoalProgress trackProgress(String userId, String goalId, boolean detailed)
+ boolean adjustPlan(String goalId, double newTarget, Date newDeadline)
IPaymentsServiceBL
+ String addBill(String userId, String provider, String accountRef, IMoney amount, Date dueAt)
+ boolean schedulePayment(String userId, String billId, Date when, boolean recurring)
+ boolean processPayment(String userId, String methodId, IMoney amount, String merchantId)
+ boolean requestRefund(String userId, String txnId, IMoney amount, String reason)
+ boolean verifyPaymentMethod(String userId, String methodId, String cvv)
IAccountService
String createAccount(String username, String email, String phoneNumber, String passwordHash, Map meta)
UserAccount getAccountById(String userId)
boolean updateAccount(String userId, String email, String phone, String address)
boolean changeUsername(String userId, String newUsername)
boolean setStatus(String userId, boolean enabled)
boolean deleteAccount(String userId)
INotificationsServiceBL
+ boolean sendAlert(String userId, String message, String channel)
+ boolean setPreference(String userId, String type, boolean enabled)
+ List<INotification> history(String userId)
+ boolean scheduleBillReminder(String userId, String billId, Date at)
IGroupSavingsService
+ String createGroup(String name, List<String> members)
+ boolean addMember(String groupId, String memberId)
+ String contribute(String groupId, String memberId, double amount, Date at)
+ IGroupReport generateReport(String groupId)
+ boolean requestWithdrawal(String groupId, String memberId, double amount)
IAuthService
AuthRepository
CredentialRecord getCredentialsByUserId(String userId)
boolean updatePassword(String userId, String passwordHash, Date updatedAt)
boolean createSession(String userId, String token, Date expiresAt, Map meta)
SessionRecord getSession(String token)
boolean revokeSession(String token)
boolean logAuthEvent(String userId, String eventType, String details, Date at)
ReportingService
BudgetService
GoalService
PaymentsServiceBL
AccountService
NotificationsServiceBL
GroupSavingsService
AuthService
IControllerOutput
void respond(string userId, string responseData)
void fail(string userId, string errorMessage)
IControllerInput
void handleRequest(string userId, string requestType, string requestData)
void handleAction(string userId, string actionType)
IControllerReports
String ctlGenerateIncomeStatement(String userId)
String ctlGenerateMonthlySummary(String userId)
boolean ctlExportToPDF(String reportType)
IControllerBudget
BudgetPlan ctlCreateBudget(String userId, double totalAmount, String startDate, String endDate, List<String> categories)
boolean ctlRecordExpense(String userId, double amount, String category, String date, String paymentMethod)
boolean ctlSendBudgetAlert(String userId, String alertType, String message)
IControllerGoals
FinancialGoal ctlSetGoal(String userId, String goalType, double targetAmount, String deadline, String priorityLevel)
GoalProgress ctlTrackProgress(String userId, String goalId, boolean detailed)
AIRecommendation ctlGetOptimizedGoalPlan(String userId, String financialData, int timeFrame, boolean riskSensitive)
IControllerPayments
boolean ctlProcessPayment(String userId, String methodId, double amount, String merchantId)
boolean ctlRequestRefund(String userId, String transactionId, double amount, String reason)
boolean ctlVerifyPaymentMethod(String userId, String methodId, String CVV)
IControllerAuth
boolean ctlCreateAccount(String username, String email, String password, String phoneNumber)
boolean ctlUpdateUserInfo(String userId, String newEmail, String newPhoneNumber, String newAddress)
boolean ctlDeleteAccount(String userId)
boolean ctlChangeUsername(String userId, String newUsername)
boolean ctlResetPassword(String userId, String oldPassword, String newPassword)
boolean ctlVerifyIdentity(String userId, String verificationCode, String method)
UserAccount ctlGetAccountDetails(String userId)
boolean ctlEnableAccount(String userId)
boolean ctlDisableAccount(String userId)
IControllerNotifications
boolean ctlSendAlert(string userId, string message)
boolean ctlSetNotificationPreference(string userId, string preferenceType, boolean isEnabled)
List<Notification> ctlGetNotificationHistory(string userId)
IControllerGroupSavings
SavingsGroup ctlCreateGroup(String groupName, List<String> members)
boolean ctlAddMember(String groupId, String memberId)
boolean ctlProcessDeposit(String groupId, String memberId, double amount)
String ctlGenerateGroupReport(String groupId)
Data Model
ReportsRepositoryImpl
GoalsRepositoryImpl
AccountsRepositoryImpl
AuthRepositoryImpl
DataAccessLayer
NotificationsRepositoryImpl
BudgetRepositoryImpl
PaymentsRepositoryImpl
GroupSavingsRepositoryImpl
ReportsRepository
List<Transaction> fetchTransactions(String userId, String period)
List<BudgetPlan> fetchBudgets(String userId, String period)
List<FinancialGoal> fetchGoals(String userId)
boolean saveExport(String userId, String type, String blobRef, Map meta)
BudgetRepository
String createBudget(String userId, double totalAmount, String startDate, String endDate, List<String> categories)
String addExpense(String userId, double amount, String category, String date, String method)
BudgetPlan getBudget(String userId, String period)
boolean updateAggregates(String userId, String period, Map deltas)
GoalsRepository
String createGoal(String userId, String type, double targetAmount, String deadline, String priority)
FinancialGoal getGoal(String goalId)
List<FinancialGoal> listGoals(String userId)
boolean writeProgress(String userId, String goalId, GoalProgress progress)
AccountsRepository
String createAccount(String username, String email, String phoneNumber, String passwordHash, Map meta)
UserAccount getAccountById(String userId)
boolean updateAccount(String userId, String email, String phone, String address)
boolean changeUsername(String userId, String newUsername)
boolean setStatus(String userId, boolean enabled)
boolean deleteAccount(String userId)
AuthRepository
CredentialRecord getCredentialsByUserId(String userId)
boolean updatePassword(String userId, String passwordHash, Date updatedAt)
boolean createSession(String userId, String token, Date expiresAt, Map meta)
SessionRecord getSession(String token)
boolean revokeSession(String token)
boolean logAuthEvent(String userId, String eventType, String details, Date at)
NotificationsRepository
boolean saveTemplate(String templateId, String content, String channel)
NotificationPreferences getPreferences(String userId)
boolean saveHistory(String userId, String messageId, String channel, String status, String at)
String scheduleMessage(String userId, String message, String channel, String at)
PaymentsRepository
String createTransaction(String userId, String methodId, double amount, String merchantId, String authRef)
boolean updateTransactionStatus(String txnId, String status, String providerRef)
PaymentMethod getPaymentMethod(String userId, String methodId)
boolean storeTokenizedCard(String userId, String token, String scheme, String last4)
GroupSavingsRepository
String createGroup(String groupName, List<String> memberIds)
boolean addMember(String groupId, String memberId)
String recordDeposit(String groupId, String memberId, double amount, String at)
GroupReportData getGroupReportData(String groupId)
«interface»
IDataAccessLayer
ResultSet query(String table, Map where, Map options)
IdRef insert(String table, Map record)
boolean update(String table, Map where, Map patch)
boolean delete(String table, Map where)
TxHandle beginTransaction()
boolean commit(TxHandle tx)
boolean rollback(TxHandle tx)
boolean publishEvent(String topic, Map payload)
ReportingRepositoryImpl
GoalRepositoryImpl
AccountRepositoryImpl
Domain Model
IBanker
IAuth
+ CredentialRecord getCredentialsByUserId(String userId)
+ boolean updatePassword(String userId, String passwordHash, Date updatedAt)
+ boolean createSession(String userId, String token, Date expiresAt, Map meta)
+ SessionRecord getSession(String token)
+ boolean revokeSession(String token)
+ boolean logAuthEvent(String userId, String eventType, String details, Date at)
INotificationPreferences
ISpendingTrends
IMoney
IPaymentTransaction
ISavingsGroup
+ String getId()
+ String getName()
+ double getTargetAmount()
IGroupReport
IReport
IGoalProgress
+ String getGoalId()
+ double getCurrentAmount()
+ double getPercentage()
+ String getStatus()
Banker
+ String id
+ String name
+ String role
+ String branch
Auth
NotificationPreferences
+ String userId
+ boolean pushEnabled
+ boolean smsEnabled
+ boolean emailEnabled
SpendingTrends
+ String userId
+ String period
+ Map<String,double> categoryTotals
Money
+ double amount
+ String currency
PaymentTransaction
+ String id
+ String userId
+ String providerRef
+ double amount
+ String status
+ Date at
SavingsGroup
+ String id
+ String name
+ double targetAmount
+ Date createdAt
GroupReport
+ String groupId
+ double totalContributed
+ double progressPercent
Report
+ String id
+ String userId
+ String type
+ String period
+ String contentRef
GoalProgress
+ String goalId
+ double currentAmount
+ double percentage
+ String status
1
1
IUserAccount
+ String getId()
+ boolean isEnabled()
+ boolean isTwoFAEnabled()
+ void enable2FA(String method)
+ void disable2FA()
1
ICustomer
many
IGroupContribution
many
IGroupMember
UserAccount
+ String id
+ String username
+ String email
+ String phone
+ String address
+ boolean enabled
+ boolean twoFAEnabled
+ Date createdAt
+ Date updatedAt
+ enable2FA(method): void
+ disable2FA(): void
Customer
+ String id
+ String fullName
+ String nationalId
+ String phone
+ String email
GroupContribution
+ String id
+ String groupId
+ String memberId
+ double amount
+ Date at
GroupMember
+ String groupId
+ String memberId
+ Date joinedAt
1
1
1
many
many
many
many
many
many
many
INotification
ITransaction
+ String getId()
+ String getCategory()
+ double getAmount()
+ Date getAt()
+ String getPaymentMethod()
IBudgetPlan
+ String getId()
+ double getTotalAmount()
+ Date getStartDate()
+ Date getEndDate()
IPaymentMethod
IBill
ILoginActivity
+ String getId()
+ Date getAt()
+ String getDevice()
+ String getIp()
+ String getStatus()
IFinancialGoal
+ String getId()
+ String getType()
+ double getTargetAmount()
+ Date getDeadline()
+ String getPriority()
Notification
+ String id
+ String userId
+ String channel
+ String message
+ String status
+ Date at
Transaction
+ String id
+ String userId
+ String category
+ double amount
+ Date at
+ String paymentMethod
BudgetPlan
+ String id
+ String userId
+ double totalAmount
+ Date startDate
+ Date endDate
PaymentMethod
+ String id
+ String userId
+ String scheme
+ String last4
+ boolean verified
Bill
+ String id
+ String userId
+ String provider
+ String accountRef
+ double amount
+ Date dueAt
+ boolean recurring
LoginActivity
+ String id
+ String userId
+ Date at
+ String device
+ String ip
+ String status
FinancialGoal
+ String id
+ String userId
+ String type
+ double targetAmount
+ Date deadline
+ String priority